The Allocation of Teaching Talent and Human Capital Accumulation

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March 17, 2026

Introduction

- Motivating Facts
  - In the U.S., resources allocated to education have increased significantly since the 1950s
  - At same time, Improvements in educational achievement have been modest by international standards
- Research Question
  - To what extent do changes in career opportunities in other occupations affect selection of workers into teaching careers?
  - To what extent are static efficiency gains associated with improved career opportunities in non-teaching occupations muted or amplified by dynamic human capital investment effects?

Stylized Facts I

Fact #1: Majority of teachers is female

Period		% Female
early 70s	Project TALENT	61.1
	Census 1980	59.8
1986-1993	NLSY79	77.7
	Census 1990	74.8
2009-2013	NLSY97	77.1
	ACS 2009-2013	76.4
2003-2004	NCES (2006)	75

Fact #2: Women face barriers in (higher) education and in labor markets

- Women face low barriers / discrimination in teaching
- Barriers / discrimination in non-teaching occupations for women generally falling over time

Stylized Facts II

Fact #3: Trends in occupational choices

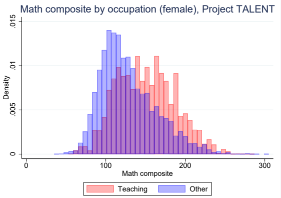
- Share of teachers in labor force rising between 1970 and 2010
  - Share of teachers among women: 3.4% in 1970, 6.0% in 2010
  - Share of teachers among men: 2.6% in 1970, 1.9% in 2010
- Sharp rise in female labor force participation rate between 1970 and 1990
- Slight decline in male labor force participation rate between 1990 and 2010

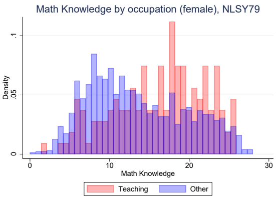
Fact #4: Trends in Education Expenditures and Scholastic Achievement

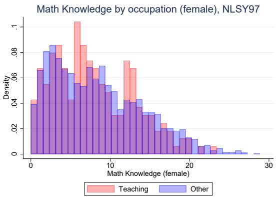
- Positive relationship between growth in spending per student and improvements in academic performance across high-income nations 1995-2015.
- Weak relationship for the United States
  (correlation 0.05 vs. 0.28 for other nations).

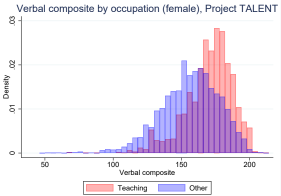
Stylized Facts III

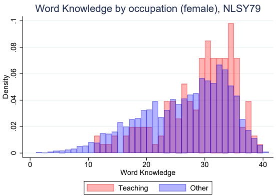
Fact #5: Trends in ability distribution of women by occupation

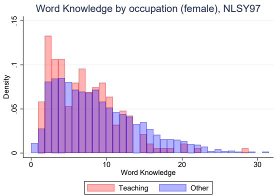












Model

Three major building blocks:

1. Two-period OLG model with human capital investment
2. Agents' occupational choices based on comparative and absolute advantage
3. Educational barriers / labor market discrimination (as in Hsieh et al., 2019)

Novel model elements:

- Human capital accumulation in classroom setting:

$$h'_{i,g}(a_i) = \underbrace{(h_{T,\hat{g}})^{\beta}}_{\text{teacher's human capital}} \overbrace{a_i^{\alpha}}^{\text{student's ability}} (s_{i,g})^{\phi} (e_{i,g})^{\eta} \underbrace{\left(N(h_{T,\hat{g}})\right)^{-\sigma}}_{\text{class size}}$$

- Total human capital in teaching $(\tilde{H}_T)$ is aggregate state variable
- Teachers' wages are funded by (local) tax revenue

Equilibrium

Given occupational choices of today's "old" and aggregate human capital in teaching $\widetilde{H}_T$, an equilibrium consists of individual choices of "young" $\{e_{T,g}, s_{T,g}, e_{O,g}, s_{O,g}\}$, the occupational choice boundary $a^*_{T,g}(a_O)$, the corresponding densities $f_{T,g}$ and $f_{O,g}$, and occupation- and group-specific wage profiles $\{\omega_{T,g}, \omega_{O,g}\}$ such that:

1. Individuals solve their investment problems

   ▸ Time Investment (s) ▸ Goods Investment (e)

2. Government budget constraint is satisfied

▸ Aggregate Law of Motion

▸ Transition Dynamics

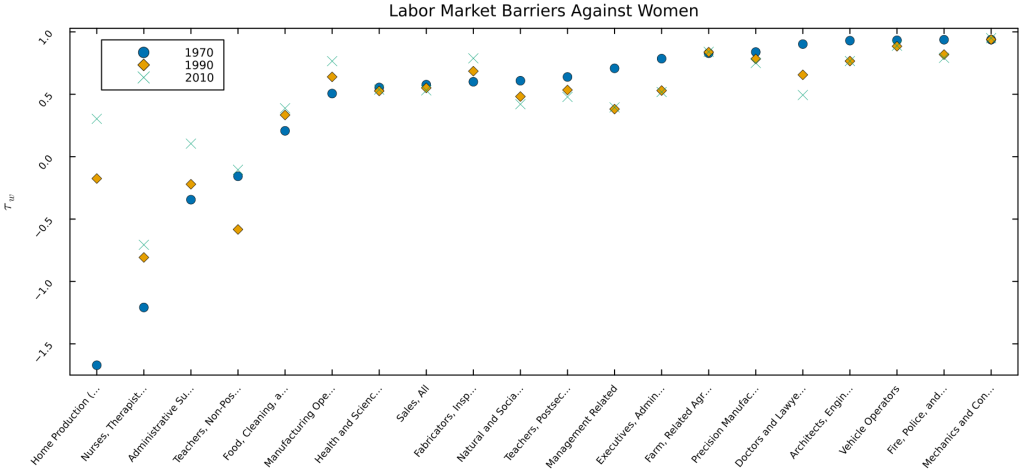
Data

- Micro-data on abilities and occupational choice:
  1. Project TALENT (1960-1975):
     - representative 5% sample of high school population in 1960
     - follow-up surveys at 1, 5, and 11-year post graduation
  2. NLSY 79
  3. NLSY 97
- Respondents' *math*, *verbal*, and *social* abilities based on aptitude tests in all three longitudinal surveys
- Occupational choice 11 years after (likely) high school graduation in all surveys ($\sim$ age 29)

Benchmark Calibration

Evolution of Labor Market Barriers for Women

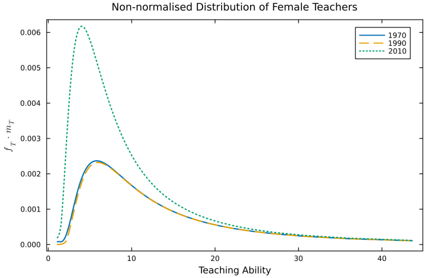
List of Occupations



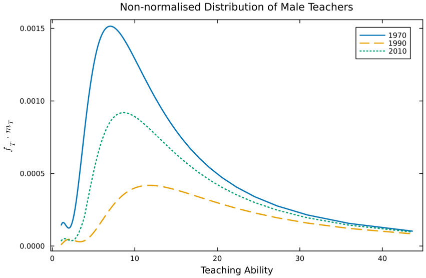
Benchmark Calibration

Distribution of Teaching Abilities

More women with lower abilities become teachers after 1990



Men exit teaching 1970-90, partially return 1990-2010



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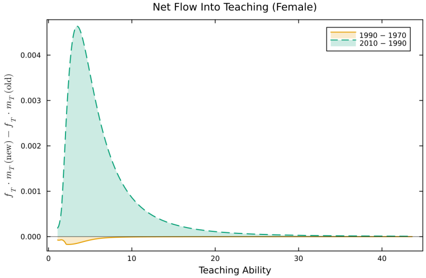

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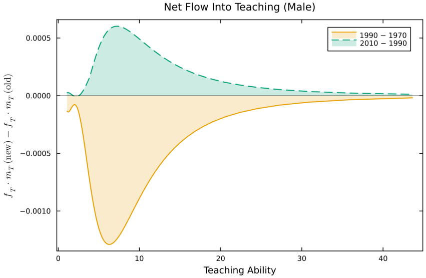


Counterfactual Experiment #1: Labor Market Barriers for Women Frozen in 1970

Comparison with Benchmark Calibration

- Number of teachers and $\widetilde{H}_T$ lower than in benchmark
- Ability distribution of female teachers more *right*-skewed and women's average h_T higher compared to benchmark
- Ability distribution of male teachers more *left*-skewed and men's average h_T lower compared to benchmark

Counterfactual Experiment #2: Barriers Except Home Production Frozen in 1970

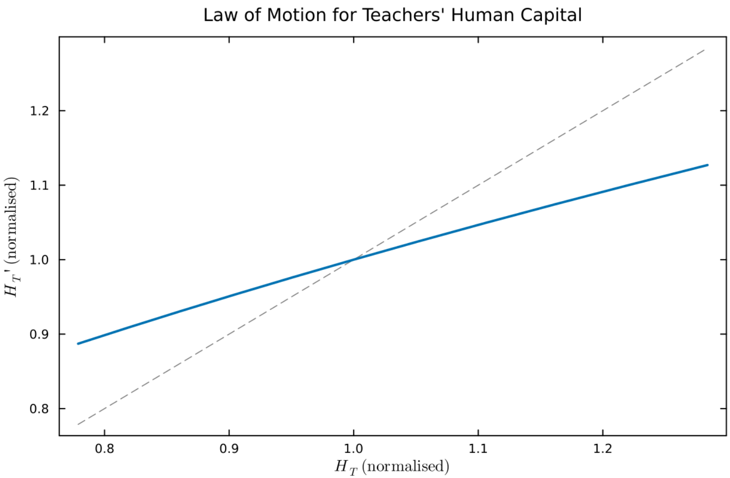
Comparison with Benchmark Calibration

- Number of teachers (male and female) and $\widetilde{H}_T$ higher than in benchmark
- Average h_T of teachers similar to benchmark
- Ability distribution of all teachers more *right*-skewed compared to benchmark

Ongoing and Future Work

- More granular decomposition of labor productivity growth into selection and human capital investment effects
- Disentangle role of $\tilde{H}_T$ and individual teachers' h_T more sharply
- Quantify extent of occupational mismatch using aptitude test scores in Project TALENT, NLSY79, NLSY97, and occupational skill requirements in O*NET
- Explore effect of competition for teaching talent across multiple locations / school districts (differentiated by amenities)

Aggregate Law of Motion



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Optimal Time Investment

$$s_{T,g} = \frac{\mu\phi}{\mu\phi + \frac{1}{\gamma} - \eta}$$

$$s_{O,g} = \frac{\mu\phi}{\mu\phi + 1 - \eta}$$

Back

Optimal Goods Investment

$$e_{T,g} = \left((\kappa \cdot \gamma \cdot \eta \cdot \tau_{T,g})^{\frac{1}{\gamma}} \cdot a_T^{\alpha} \cdot s_{T,g}^{\phi} \cdot \left(\frac{2\widetilde{H}_T}{M} \right)^{\sigma} \right)^{\frac{1}{\frac{1}{\gamma} - \eta}}$$

$$e_{O,g} = \left(A'_O \cdot \eta \cdot \tau_{O,g} \cdot a_O^{\alpha} \cdot s_{O,g}^{\phi} \cdot \left(\frac{2\widetilde{H}_T}{M} \right)^{\sigma} \right)^{\frac{1}{1-\eta}}$$

where

$$\tau_{i,g} = \frac{(1-t)(1-\tau_{i,g}^{\omega})}{1+\tau_{i,g}^{e}}$$

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Aggregate Law of Motion

$$\widetilde{H}'_T = \left[(\kappa \cdot \gamma \cdot \eta)^{\frac{\eta}{1-\eta\gamma}} \cdot \left(\frac{2\widetilde{H}_T}{M} \right)^{\frac{\sigma}{1-\eta\gamma}} \times \sum_{g=1}^{G} \tau_{T,g}^{\frac{\eta}{1-\eta\gamma}} \cdot \int_0^\infty s_{T,g}^{\frac{\phi}{1-\eta\gamma}} \cdot a^{\frac{\alpha}{1-\eta\gamma}} f_{T,g}(a) da \right]^{\frac{\beta}{\sigma}}$$

where

$$\tau_{i,g} = \frac{(1-t)(1-\tau_{i,g}^{\omega})}{1+\tau_{i,g}^{e}}$$

Back

Transition Dynamics

- Occupational choice boundary depends on $\widetilde{H}_T$:

$$\frac{a^*_{T,g}(a_O)^{\frac{\alpha}{\frac{1}{\gamma}-\eta}}}{a_O^{\frac{\alpha}{1-\eta}}} \cdot \frac{s_{T,g}^{\frac{\phi}{\frac{1}{\gamma}-\eta}}}{s_{O,g}^{\frac{\phi}{1-\eta}}} \cdot \frac{\tau_{T,g}^{\frac{1}{1-\eta\gamma}}}{\tau_{O,g}^{\frac{1}{1-\eta}}} \cdot \frac{1+\tau^e_{T,g}}{1+\tau^e_{O,g}} \cdot \left(\frac{1-s_{T,g}}{1-s_{O,g}}\right)^{\frac{1}{\mu}}$$

$$\times \frac{(\kappa \cdot \gamma)^{\frac{1}{1-\eta\gamma}}}{A'^{\frac{1}{1-\eta}}_O} \cdot \frac{\frac{1}{\gamma}-\eta}{1-\eta} \cdot \eta^{\frac{\eta(\gamma-1)}{(1-\eta)(1-\eta\gamma)}} \cdot \left(\frac{2\widetilde{H}_T}{M}\right)^{\frac{\sigma(\gamma-1)}{(1-\eta)(1-\eta\gamma)}} = 1$$

where

$$\tau_{i,g} = \frac{(1-t)(1-\tau^\omega_{i,g})}{1+\tau^e_{i,g}}$$

- Relationship between $\widetilde{H}_T$ and $a^*_{T,g}(a_O)$ "drives" transition dynamics

List of Occupations

1. Executives, Administrative, and Managerial
2. Management Related
3. Architects, Engineers, Math, and Computer Science
4. Natural and Social Scientists, Recreation, Religious, Arts, Athletes
5. Doctors and Lawyers
6. Nurses, Therapists, and Other Health Service
7. Teachers, Postsecondary
8. Teachers, Non-Postsecondary and Librarians
9. Health and Science Technicians
10. Sales, All
11. Administrative Support, Clerks, Record
12. Fire, Police, and Guards
13. Food, Cleaning, and Personal Services and Private Household
14. Farm, Related Agriculture, Logging, and Extraction
15. Mechanics and Construction
16. Precision Manufacturing
17. Manufacturing Operators
18. Fabricators, Inspectors, and Material Handlers
19. Vehicle Operators
20. Home Production

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